

Primary 5 Science Practice Sheets

5 Sets - MOE-style MCQ Revision

Name: _____ Class: _____ Date: _____

Instructions: For each question, choose the best answer and write its number in the brackets.

This worksheet uses original questions modelled on Primary 5 Science concepts.

Topics: heat, light, magnets, matter, water cycle, life cycles, reproduction, materials and fair tests.

Layout: Maximum two questions per page. Answer sheet is placed after each set.

Set 1 - Life Cycles, Reproduction, Heat and Matter

Choose the best answer and write the option number in the brackets.

1. Which characteristic is found in birds but not in other animals?

- (1) They can fly.
- (2) They lay eggs.
- (3) They have two wings.
- (4) They have feathers on their bodies.

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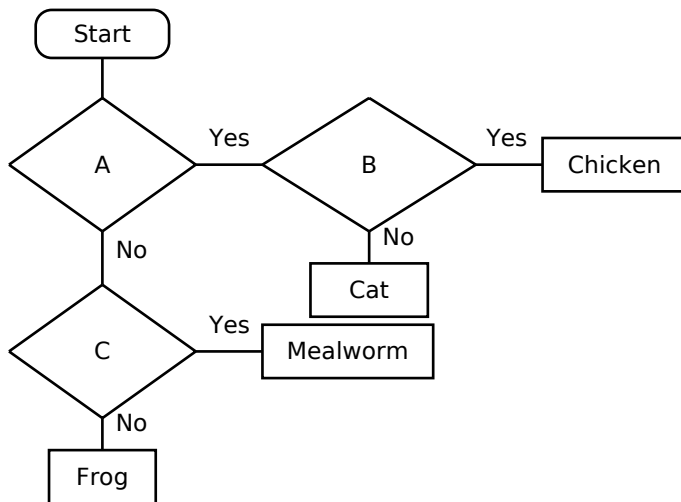
2. A mouldy orange and bacteria in spoiled food are both living things. Which statement about them is true?

- (1) Both are microorganisms.
- (2) Both make their own food.
- (3) Both reproduce by seeds.
- (4) Both are flowering plants.

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Set 1 - Life Cycles, Reproduction, Heat and Matter

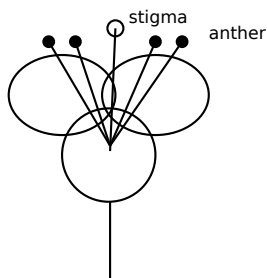
3. Study the flowchart on the life cycles of a chicken, cat, mealworm and frog. Which row best represents A, B and C?



- (1) A: Has a 3-stage life cycle; B: Gives birth alive; C: Young resembles adult
 (2) A: Young resembles adult; B: Lays eggs; C: Has a 4-stage life cycle
 (3) A: Has a 4-stage life cycle; B: Young resembles adult; C: Lays eggs
 (4) A: Gives birth alive; B: Has a 3-stage life cycle; C: Young resembles adult

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4. Flower X is pollinated by wind. Which features help wind pollination?

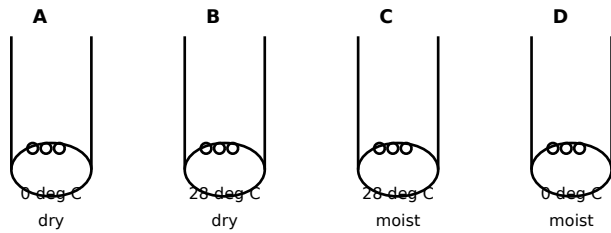


- (1) Large colourful petals only.
 (2) Feathery stigma and anthers hanging outside the flower.
 (3) Sweet smell and nectar.
 (4) Sticky pollen and small stigma only.

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Set 1 - Life Cycles, Reproduction, Heat and Matter

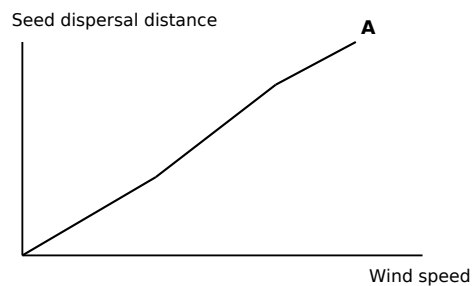
5. Aishah wants to find out if warmth affects the rate of germination of bean seeds. Which two set-ups should she compare?



- (1) A and B
- (2) A and C
- (3) B and D
- (4) C and D

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6. Seeds X, Y and Z have wing-like structures of different sizes. Line A shows the greatest distance travelled as wind speed increases. Which seed most likely represents Line A?



- (1) Seed with the smallest wing area.
- (2) Seed with no wing-like structure.
- (3) Seed with the largest wing area.
- (4) Seed with the heaviest mass only.

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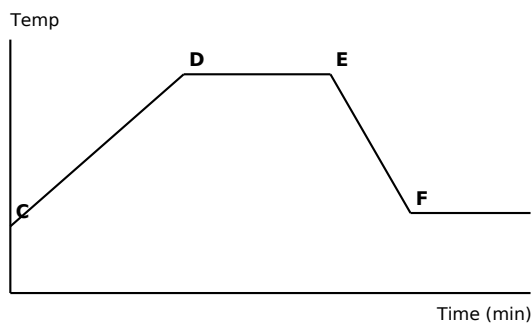
Set 1 - Life Cycles, Reproduction, Heat and Matter

7. During fertilisation in humans, which statement is correct?

- (1) Many sperm cells may reach the egg, but only one fertilises it.
- (2) The fertilised egg develops in the stomach.
- (3) The fertilised egg contains genetic information only from the father.
- (4) Fertilisation happens outside the female body.

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8. The graph shows changes in the temperature of water over time. At the flat part from D to E, what is happening?

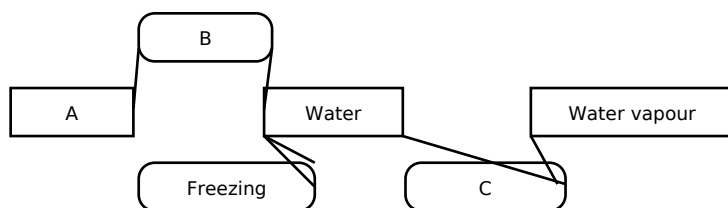


- (1) The water is only liquid.
- (2) The water is only gas.
- (3) The water is changing state and contains liquid and gas.
- (4) The water is changing from liquid to solid.

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Set 1 - Life Cycles, Reproduction, Heat and Matter

9. The diagram shows states of water and some processes. What are A, B and C?



- (1) A: Ice; B: Melting; C: Condensation
- (2) A: Ice; B: Boiling; C: Evaporation
- (3) A: Steam; B: Melting; C: Condensation
- (4) A: Steam; B: Boiling; C: Evaporation

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10. Janet wants to find out whether temperature affects the rate of evaporation of water. Which two set-ups should she use?

Set-up	Surface area	Water vol.	Temp	Wind
A	150 cm ²	200 ml	28 deg C	Absent
B	60 cm ²	300 ml	35 deg C	Present
C	150 cm ²	200 ml	35 deg C	Absent
D	60 cm ²	300 ml	28 deg C	Absent

- (1) A and B
- (2) A and C
- (3) B and D
- (4) C and D

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Set 1 - Life Cycles, Reproduction, Heat and Matter - Answer Sheet

Answer Sheet

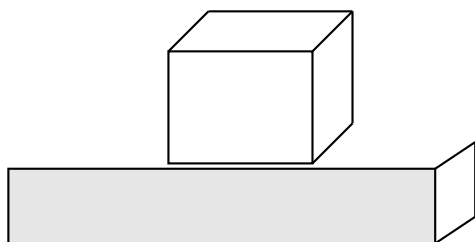
Answers are shown with the main concept tested. No working space is required on this page.

Q	Answer	Key idea
1	(4)	Birds have feathers.
2	(1)	Bacteria and mould are microorganisms.
3	(2)	Use yes/no traits to classify life cycles.
4	(2)	Wind-pollinated flowers expose anthers and have feathery stigmas.
5	(4)	Change temperature only; keep water/moisture the same.
6	(3)	Larger wing area helps wind dispersal farther.
7	(1)	One sperm fertilises the egg; embryo develops in uterus.
8	(3)	Temperature stays constant during boiling while state changes.
9	(1)	Ice melts to water; water vapour condenses to water.
10	(2)	Fair test: only temperature changes.

Set 2 - Heat, Light, Magnets and Materials

Choose the best answer and write the option number in the brackets.

1. Alice placed an ice cube on a metal block in a room. After thirty minutes, which description is correct?



- (1) The metal block decreased in temperature because it lost heat to the ice cube.
- (2) The metal block increased in temperature because it gained heat from the ice cube.
- (3) The metal block decreased in temperature because the ice cube lost heat to it.
- (4) The metal block increased in temperature because the ice cube gained heat from it.

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2. Jenny wrapped two identical cans of hot water with metal strips. Can A had a larger area of metal touching the can. Which conclusion is correct?

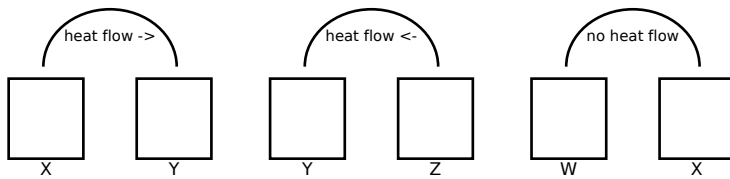
Time	Can A	Can B
0 min	50 deg C	50 deg C
5 min	44 deg C	47 deg C
10 min	40 deg C	44 deg C
15 min	36 deg C	42 deg C

- (1) Can A lost heat faster because more metal was in contact with the can.
- (2) Can B lost heat faster because less metal was in contact with the can.
- (3) Can A lost heat faster because its metal was a poorer conductor.
- (4) Can B lost heat faster because its metal was a better conductor.

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Set 2 - Heat, Light, Magnets and Materials

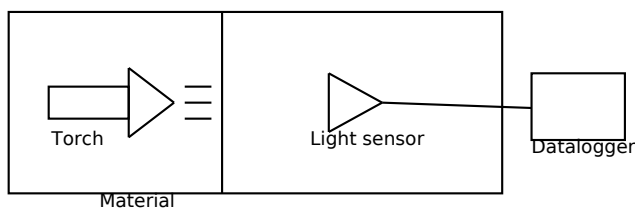
3. The arrows show the direction of heat flow between beakers connected by metal strips. Which statement is correct?



- (1) X is hotter than Y.
- (2) Z is hotter than Y.
- (3) W and X have the same temperature.
- (4) Y is colder than X and hotter than Z.

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4. Sarah shone a torch at three materials and measured the light detected. X=300 units, Y=80 units, Z=0 units. Which set best represents X, Y and Z?

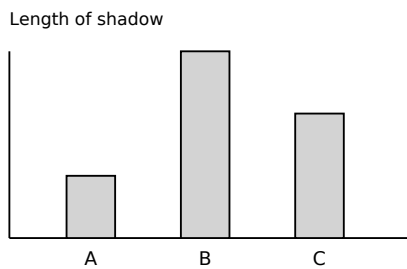


- (1) X: frosted glass; Y: mirror; Z: cardboard
- (2) X: clear plastic; Y: frosted glass; Z: cardboard
- (3) X: mirror; Y: clear plastic; Z: cardboard
- (4) X: cardboard; Y: frosted glass; Z: clear plastic

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Set 2 - Heat, Light, Magnets and Materials

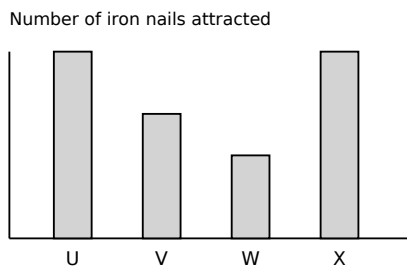
5. Three identical wooden blocks are placed at different distances from a light source in front of a screen. Block B is nearest the light source, A is farthest and C is in between. Which shadow pattern is correct?



- (1) A has the longest shadow, then B, then C.
- (2) B has the longest shadow, then C, then A.
- (3) C has the longest shadow, then A, then B.
- (4) All shadows have the same length.

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6. A bar magnet is placed in a tray of iron nails. Which graph correctly shows the number of nails attracted at U, V, W and X?

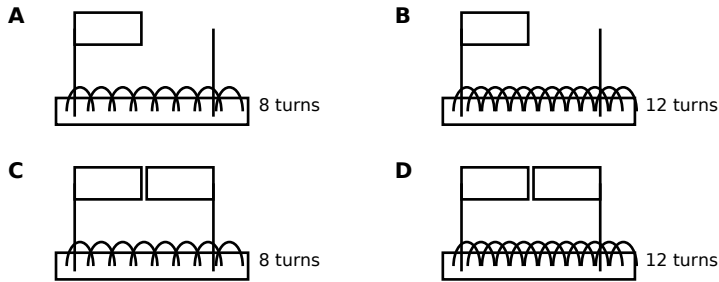


- (1) All positions attract the same number of nails.
- (2) U and X attract the most nails; W attracts the least.
- (3) U attracts the most and X attracts the least.
- (4) The number of nails increases steadily from U to X.

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Set 2 - Heat, Light, Magnets and Materials

7. Which two set-ups should be compared to find out whether the number of coils affects the strength of an electromagnet?



- (1) A and B
- (2) A and C
- (3) B and C
- (4) C and D

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8. A plastic rod has three rings. Ring A rests on the base, ring B touches A, and ring C floats above B without touching it. Which is the most likely arrangement?

- (1) A: magnet, B: magnet, C: iron ring
- (2) A: iron ring, B: iron ring, C: magnet
- (3) A: iron ring, B: magnet, C: magnet
- (4) A: magnet, B: iron ring, C: iron ring

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Set 2 - Heat, Light, Magnets and Materials

9. A swimming float must help a person stay afloat in water. Which material is most suitable?

Material	Flexible	Waterproof	Floats	Transparent
A	No	Yes	Yes	Yes
B	Yes	No	Yes	No
C	Yes	Yes	Yes	No
D	No	Yes	Yes	No

- (1) A
- (2) B
- (3) C
- (4) D

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10. Daniel pushed the plunger of a syringe containing sand and air. The tip was blocked. What happens?

- (1) Volume of sand unchanged; volume of air unchanged.
- (2) Volume of sand decreased; volume of air decreased.
- (3) Volume of sand unchanged; volume of air decreased.
- (4) Volume of sand increased; volume of air unchanged.

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Set 2 - Heat, Light, Magnets and Materials - Answer Sheet

Answer Sheet

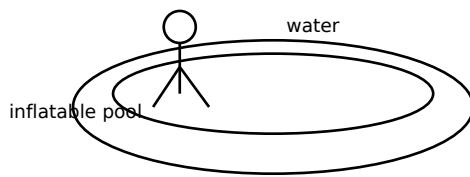
Answers are shown with the main concept tested. No working space is required on this page.

Q	Answer	Key idea
1	(1)	Heat travels from warmer metal to colder ice.
2	(1)	More conducting surface area increases heat loss.
3	(3)	Heat flows from hotter to colder; no flow means same temperature.
4	(2)	Transparent lets most light through; translucent some; opaque none.
5	(2)	An object nearer the light source forms a larger shadow.
6	(2)	Magnetic force is strongest at the poles.
7	(1)	Change only the number of coils; keep battery number the same.
8	(3)	Two magnets can repel and create a gap; iron is attracted to magnet.
9	(3)	A float should be flexible, waterproof and able to float.
10	(3)	Solids cannot be compressed easily; gases can be compressed.

Set 3 - Matter, Water Cycle, Evaporation and Reproduction

Choose the best answer and write the option number in the brackets.

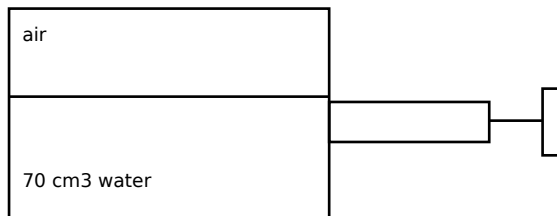
1. An inflatable pool is filled to the brim with water. When Katy enters the pool, water spills out. Which property of matter causes this?



- (1) Water has mass.
- (2) Katy occupies space.
- (3) Katy has a definite shape.
- (4) The pool cannot be compressed.

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2. An air-tight container has a volume of 100 cm³. It contains 70 cm³ of water. Amanda removes 10 cm³ of water and pumps in 20 cm³ of air. What is the final volume of air in the container?



- (1) 20 cm³
- (2) 30 cm³
- (3) 40 cm³
- (4) 50 cm³

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Set 3 - Matter, Water Cycle, Evaporation and Reproduction

3. Study the table. Which conclusion is correct?

Substance	15 deg C	35 deg C	55 deg C
A	Liquid	Liquid	Gas
B	Liquid	Liquid	Gas
C	Liquid	Liquid	Liquid

- (1) The boiling point of A is exactly 55 deg C.
- (2) The melting point of B is 15 deg C.
- (3) Substance C has the highest boiling point.
- (4) Substance B has a lower melting point than C.

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4. A kettle of water was boiled and a white cloud X was seen near the spout. Which statements are correct? A: X and clouds are formed by condensation. B: X and clouds are made of water droplets. C: X is steam while clouds are water vapour. D: X is in gaseous state while clouds are liquid.

- (1) C only
- (2) A and B only
- (3) B and D only
- (4) A, C and D only

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Set 3 - Matter, Water Cycle, Evaporation and Reproduction

5. Ian placed two identical beakers, C and D, in the same basin of warm water. Both started with 100 ml of water. After some time, C had 85 ml left and D had 92 ml left. If D had an exposed surface area of 30 cm², what could be the exposed surface area of C?

- (1) 20 cm²
- (2) 30 cm²
- (3) 40 cm²
- (4) 10 cm²

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6. In humans, which statement about reproductive cells is correct?

- (1) The male reproductive cell is produced in the testes.
- (2) The female reproductive cell is produced in the uterus.
- (3) The fertilised egg develops in the stomach.
- (4) The baby receives genetic information only from the mother.

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Set 3 - Matter, Water Cycle, Evaporation and Reproduction

7. A flower has its petals removed before pollination. Which statement is most likely correct?

- (1) It cannot undergo fertilisation because petals produce pollen.
- (2) It may still undergo fertilisation if pollen reaches its stigma.
- (3) It cannot make seeds because petals are the female part.
- (4) It can reproduce only if it has colourful petals.

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8. Which pair correctly matches a process to a change of state?

- (1) Melting: liquid to solid
- (2) Evaporation: gas to liquid
- (3) Condensation: gas to liquid
- (4) Freezing: solid to liquid

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Set 3 - Matter, Water Cycle, Evaporation and Reproduction

9. A student places a hot metal spoon and a wooden spoon in hot soup for 3 minutes. Which observation and explanation is correct?

- (1) The metal spoon feels hotter because metal conducts heat better.
- (2) The wooden spoon feels hotter because wood conducts heat better.
- (3) Both spoons feel equally hot because they are in the same soup.
- (4) The metal spoon feels colder because it loses heat faster.

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10. A compass is placed near the north pole of a bar magnet. What happens to the north-seeking end of the compass needle?

- (1) It points towards the north pole of the bar magnet.
- (2) It points away from the north pole of the bar magnet.
- (3) It spins continuously.
- (4) It will not move because a compass is not magnetic.

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Set 3 - Matter, Water Cycle, Evaporation and Reproduction - Answer Sheet

Answer Sheet

Answers are shown with the main concept tested. No working space is required on this page.

Q	Answer	Key idea
1	(2)	Matter occupies space.
2	(3)	Container volume is fixed; air occupies the space not filled by water.
3	(3)	C has not boiled at 55 deg C, so its boiling point is higher.
4	(2)	Visible white cloud is tiny water droplets from condensation.
5	(3)	A larger exposed surface area increases the rate of evaporation.
6	(1)	Sperm are produced in testes; eggs in ovaries.
7	(2)	Petals attract pollinators; reproductive parts are needed for fertilisation.
8	(3)	Condensation changes water vapour to liquid water.
9	(1)	Metals are good conductors of heat.
10	(2)	Like poles repel; the compass north end is repelled by a north pole.

Set 4 - Classification, Fair Tests and Scientific Reasoning

Choose the best answer and write the option number in the brackets.

1. Which animal has a 4-stage life cycle?

- (1) Chicken
- (2) Cat
- (3) Frog
- (4) Mealworm beetle

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2. Which comparison between a frog and a chicken is correct?

- (1) Both have young that look like the adult.
- (2) Both lay eggs, but the frog undergoes metamorphosis.
- (3) Both give birth to young alive.
- (4) Both have a 4-stage life cycle.

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Set 4 - Classification, Fair Tests and Scientific Reasoning

3. Why must pollen be transferred to the stigma before fertilisation in flowering plants?

- (1) The pollen contains the male reproductive cell.
- (2) The stigma produces seeds directly.
- (3) The petal changes into a fruit.
- (4) The ovary produces pollen grains.

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4. A student releases three winged seeds from the same height. Which factor should be changed to test whether wing area affects distance travelled?

- (1) The height of release
- (2) The mass and shape together
- (3) The area of the wing-like structure only
- (4) The type of fan and the room size

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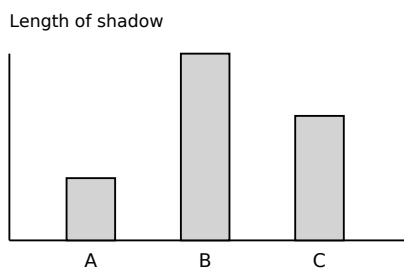
Set 4 - Classification, Fair Tests and Scientific Reasoning

5. To test if the number of batteries affects the strength of an electromagnet, which variables should be kept the same?

- (1) Number of coils and type of iron core
- (2) Number of batteries and number of paper clips
- (3) Number of coils only
- (4) Number of batteries only

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6. A shadow becomes larger when an object moves closer to a light source. Which graph best shows block B nearest, C in the middle and A farthest?



- (1) B tallest, C medium, A shortest
- (2) A tallest, B medium, C shortest
- (3) All bars equal
- (4) C tallest, A medium, B shortest

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Set 4 - Classification, Fair Tests and Scientific Reasoning

7. Which statement about magnets is correct?

- (1) A magnet attracts all metals.
- (2) A magnet has the strongest magnetic force at its centre.
- (3) A magnet can attract iron without touching it.
- (4) A magnet has only one pole.

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8. A raincoat should be made from a material with which properties?

- (1) Flexible, waterproof and light
- (2) Transparent, hard and magnetic
- (3) Absorbent and rough
- (4) Brittle and heavy

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Set 4 - Classification, Fair Tests and Scientific Reasoning

9. A hot drink cools faster in a metal cup than in a plastic cup. Which explanation is best?

- (1) Metal is a better conductor of heat than plastic.
- (2) Plastic is always colder than metal.
- (3) Metal is waterproof but plastic is not.
- (4) Plastic makes heat disappear.

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10. A mirror is placed in the path of a beam of light. What property of the mirror explains the change in direction of the light?

- (1) It is transparent.
- (2) It reflects light.
- (3) It absorbs all light.
- (4) It allows light to pass through.

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Set 4 - Classification, Fair Tests and Scientific Reasoning - Answer Sheet

Answer Sheet

Answers are shown with the main concept tested. No working space is required on this page.

Q	Answer	Key idea
1	(4)	Mealworm beetle has egg, larva, pupa, adult.
2	(2)	Frog changes form during its life cycle; chicken lays eggs but young resembles adult.
3	(1)	Pollen carries male reproductive cells.
4	(3)	A fair test changes one variable only.
5	(1)	Keep all other variables constant.
6	(1)	Nearer objects cast larger shadows.
7	(3)	Magnetic force can act at a distance and attracts magnetic materials.
8	(1)	Materials are selected based on useful properties.
9	(1)	Good conductors transfer heat faster.
10	(2)	Mirrors reflect light.

Set 5 - Mixed Revision: MOE-style MCQ Practice

Choose the best answer and write the option number in the brackets.

- 1.** Which pair of materials would be best to compare if you want to find out whether light can pass through different materials?

- (1) A clear plastic sheet and a frosted glass sheet
- (2) A magnet and an iron nail
- (3) A metal spoon and a wooden spoon
- (4) Two identical clear plastic sheets

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- 2.** A student places a cube of ice on a warm metal tray. Which direction does heat travel?

- (1) From the ice to the metal tray
- (2) From the metal tray to the ice
- (3) From the surrounding air only to the ice
- (4) Heat does not travel between them

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Set 5 - Mixed Revision: MOE-style MCQ Practice

3. Which set-up should be used to test if water is needed for seed germination?

- (1) Seeds on dry cotton wool and seeds on moist cotton wool at the same temperature
- (2) Seeds at 0 deg C and seeds at 28 deg C, both dry
- (3) Seeds in the dark and seeds in light, both dry
- (4) Different types of seeds in moist cotton wool

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4. A student observes that a seed with a larger wing-like structure travels farther in the wind. What is the best conclusion?

- (1) All seeds travel the same distance in wind.
- (2) A larger wing-like structure helps seed dispersal by wind.
- (3) The seed travelled farther because it was heavier.
- (4) Wind has no effect on seed dispersal.

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Set 5 - Mixed Revision: MOE-style MCQ Practice

5. Which statement about bacteria is correct at Primary 5 level?

- (1) They are visible clearly with the naked eye.
- (2) They are microorganisms.
- (3) They are flowering plants.
- (4) They reproduce by laying eggs.

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6. A student uses the same electromagnet but increases the number of coils around the iron nail. What is likely to happen?

- (1) The electromagnet becomes weaker.
- (2) The electromagnet becomes stronger.
- (3) The electromagnet stops working.
- (4) The number of batteries decreases automatically.

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Set 5 - Mixed Revision: MOE-style MCQ Practice

7. Which observation shows that air can be compressed?

- (1) Air takes the shape of a balloon.
- (2) The volume of trapped air in a syringe decreases when the plunger is pushed.
- (3) Air has mass.
- (4) Air cannot be seen.

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8. Which process happens when water droplets form on the outside of a cold drink can?

- (1) Melting
- (2) Freezing
- (3) Evaporation
- (4) Condensation

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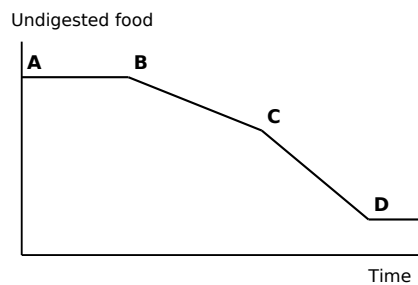
Set 5 - Mixed Revision: MOE-style MCQ Practice

9. A flower has its anthers removed but its stigma and ovary are still present. Which statement is most reasonable?

- (1) It may still be fertilised by pollen from another flower.
- (2) It cannot receive pollen because the stigma is missing.
- (3) It cannot form seeds because petals are missing.
- (4) It has become a male flower only.

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10. Undigested food moves from the small intestine to the large intestine. What generally happens to the amount of undigested food as time passes through the digestive system?



- (1) It increases continuously.
- (2) It decreases as digestion and absorption occur, then remains as waste.
- (3) It stays exactly the same at all points.
- (4) It becomes water vapour.

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Set 5 - Mixed Revision: MOE-style MCQ Practice - Answer Sheet

Answer Sheet

Answers are shown with the main concept tested. No working space is required on this page.

Q	Answer	Key idea
1	(1)	Compare different materials when testing light transmission.
2	(2)	Heat travels from warmer object to colder object.
3	(1)	Change moisture only.
4	(2)	Wing-like structures increase wind dispersal distance.
5	(2)	Bacteria are microorganisms.
6	(2)	More coils usually make an electromagnet stronger.
7	(2)	Gases can be compressed.
8	(4)	Water vapour condenses on cold surfaces.
9	(1)	Pollen from another flower can fertilise if female parts remain.
10	(2)	Food is digested/absorbed; remaining waste enters large intestine.